

Writing Inequalities from Verbal Constraints

Turning a comparison phrase into a symbol, building a one-variable inequality from a story, and solving it and reading the answer back into the situation

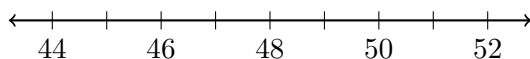
The four phrases that decide the symbol.

Phrase	Symbol	Is the boundary allowed?
more than, over	$>$	no — open circle on the number line
less than, under	$<$	no — open circle
at least, no less than, minimum	$\geq$	yes — closed circle
at most, no more than, maximum	$\leq$	yes — closed circle

Directions.

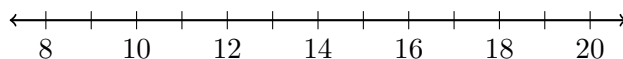
- Say in words what the variable counts or measures before you write anything — that sentence is part of the answer.
- Write the inequality first, then solve it. Graph the solution on the number line under the problem, using an open or closed circle to match your symbol.
- If a step multiplies or divides both sides by a negative number, the inequality symbol flips.

1. A ride at the fair requires riders to be *at least* 48 inches tall. Let h be a rider's height in inches. Write an inequality for the heights that may ride, and graph it.



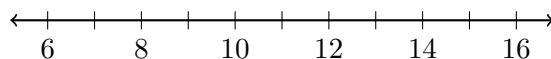
Answer: _____

2. A school van can carry *no more than* 14 passengers. Let p be the number of passengers on a trip. Write an inequality for the trips that are allowed, and graph it.



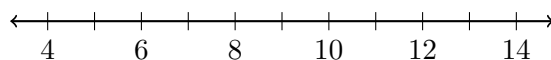
Answer: _____

3. Maya has \$60 to spend at the fair. Entry costs \$16 and each snack costs \$4. Let s be the number of snacks she buys. Write an inequality saying she does not overspend, then solve it and graph the solution.



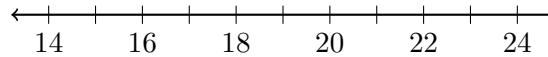
Answer: _____

4. A gym charges a \$30 joining fee once, then \$12 for each month of membership. Deshawn can spend *at most* \$150 in total. Let m be the number of months. Write an inequality, solve it, and graph the solution.



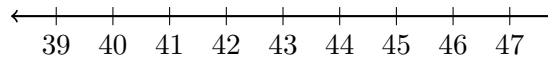
Answer: _____

- 5.** A club needs to raise *at least* \$500. They already have \$180 and sell wreaths for \$16 each. Let w be the number of wreaths sold. Write an inequality for reaching the goal, solve it, and graph the solution.



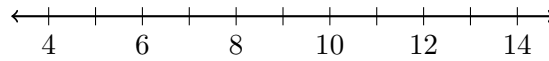
Answer: _____

- 6.** A freight elevator may carry *no more than* 2100 pounds. The operator weighs 165 pounds and each box weighs 45 pounds. Let b be the number of boxes loaded. Write an inequality, solve it, graph the solution, and then say how many boxes the operator may actually load.



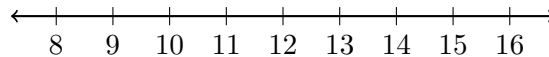
Answer: _____

7. A drone starts a descent from a height of 240 metres and falls 15 metres every second. Let t be the number of seconds since the descent began. Write an inequality for the times when the drone is *above* 90 metres, solve it, and graph the solution. Watch what happens to the symbol when you divide.



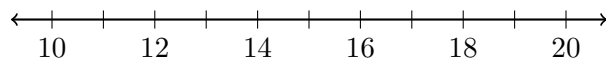
Answer: _____

8. Jonah is told “a number n is no more than 12” and writes $n < 12$. His answer is not quite right. Explain in one sentence what is wrong with his symbol, write the correct inequality, and graph it.



Answer: _____

9. A rectangle has a width of 8 centimetres and a length of l centimetres. Its perimeter must be *at most* 46 centimetres. Write an inequality for the length, solve it, and graph the solution.



Answer: _____

10. Now work backwards. Invent a real situation that the inequality $25 + 8n \geq 65$ could describe. Say clearly what n counts and what the two numbers mean in your story, then explain what the solution $n \geq 5$ tells somebody in that situation. Graph $n \geq 5$ on the line below.

